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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 11/01/2005 |
| **PI Name:** | Tosini, Gianluca Ph.D. | | |
| **Project Title:** | Preventing Desynchronization of the Circadian System in Long-Term Space Flight | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Atlanta | **State:** | GA |
| **Zip Code:** | 30310-1495 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2003 Biomedical Research & Countermeasures 03-OBPR-04 |
| **Start Date:** | 09/01/2004 | **End Date:** | 08/31/2007 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Fukuhara, Chiaki ( Morehouse School of Medicine ) | | |
| **Grant/Contract No.:** | NCC 9-58-HPF00407 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Many biochemical, physiological and behavioral parameters exhibited by organisms show daily fluctuations and most of these daily rhythms persist in constant conditions, thus, demonstrating that they are driven by endogenous oscillators. The rhythms that persist in constant conditions with a period close to 24 hours are called circadian rhythms. One the most important aspects of space flight is the absence of geophysical 24-h cycles, which, of course, affects the overall temporal organization of the organisms. In the case of long-duration manned space flight, it is crucial to understand how the whole circadian system would react and behave in such circumstances. We discovered that exposing rats to constant dim light for 60 days induce internal desynchronization in a 20-30 % of the rats tested. In the present research proposal we plan to further investigate this phenomenon, and to test specific countermeasures to obviate the occurrence of internal desynchronization in animals exposed to constant dim light. In particular, we will test if melatonin injection and exposure to brief pulse (skeleton photoperiod) of light (white light and light of different wavelength) can prevent | | |

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| | internal desynchronization. Internal desynchronization has profound effects on the capability of the organisms to perform (mentally and physically) and to remain healthy. In this research proposal, we have designed a series of experiment aimed to understand the mechanisms that are responsible for the observed desynchronization and to test specific countermeasures aimed to prevent internal desynchronization. We believe that the model we have generated will be useful in to foreseeing and preventing dysfunction of the circadian system that may arise after long periods in the space environment where the normal cycle has been altered. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | The experiments that we have carried out during this first year have provided important new data about the functional organization of the mammalian circadian system. As proposed in the original application, we have terminated the two experiments proposed in specific aim 1 (to investigate the physiological consequences of pineal Aa-nat mRNA desynchronization) and we have already collected the brains to perform the experiment proposed in Specific aim 2 (to investigate the cellular and molecular mechanisms responsible for the desynchronization between locomotor activity pattern and melatonin). As proposed in our research plan during the second year of funding, we will start the experiments described in Specific aim 3 (to investigate the use of melatonin and light (skeleton photoperiod) as effective countermeasures to prevent desynchronization). |
| **Bibliography Type:** | Description: (Last Updated: 02/21/2008) |
| **Articles in Peer-reviewed Journals** | Fukuhara, C. Aguzzi, J. Bullock, N. Tosini, G. "Effect of Long-Term Exposure to Constant Dim Light on the Circadian System of Rats" N/A , Jan-2005 |
| **Presentation** | Aguzzi, J. Bullock N., Tosini, G. "Preventing Desynchronization of the Circadian System" N/A<br>Jan-2005 |
| **Presentation** | Bullock, N., Aguzzi, J., Tosini, G. "Long-term exposure to Constant Red Dim Light Desynchronizes the Circadian System of Rats" N/A<br>Jun-2005 |
| **Presentation** | Tosini, G. "A Hierarchical Network of clocks regulates Circadian Rhythms in the Mammalian Retina" N/A<br>Jun-2005 |